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September 21, 1972
SUPERSEDING
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May 21, 1964

FEDERAL SPECIFICATION

LEAD REPOINTER, PENCIL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mechanical lead repointers designed for one-hand use in pointing plastic and graphite lead which projects from mechanical holder or wood casings.

1.2 Classification.

1.2.1 Types.

Type I - Abrasive Cup.
Type II - Metal Cutter.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington DC, Atlanta, Chicago, Kansas City MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

E18 - Rockwell Hardness - Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N. W., Washington DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The material shall be of good quality, suitable for each component and its functions.

3.2 Design. The design of the repointer shall permit one-hand operation and repoint graphite and plastic lead by revolving the head with the lead holder. Various size adapters, lead point cleaners and lead length gauges shall be furnished if required for efficient operation of the repointer. The lead holder aperture shall accommodate standard two millimeter diameter lead with lead holder aperture being a minimum 0.345-inch diameter.

3.2.1 Type I. Repointers shall point lead by rubbing the lead against an abrasive. The abrasive medium shall be readily replaceable when it becomes ineffective without the use of cements. The abrasive cups shall be furnished in a fine or coarse grit of the grade furnished commercially. (See 6.2.) Emptying of the residue or changing cups shall be accomplished without tools. The repointer shall be of sufficient weight as not to require a fixed mounting for one-hand operations. The base shall have a permanent resilient padding attached to the bottom sufficiently thick to prevent damage to surfaces upon which it is placed.

3.2.2 Type II. Repointers shall point the lead by rotating against a cutting blade or a drum with teeth which will produce a finely cut precise point. The blade of teeth may be made of tungsten carbide or tool steel. Tool steel drums or blades shall have a Rockwell hardness of C-54 to C-62. (See 4.4.1.) Means shall be provided to permit emptying the residue and accessibility to the pointing mechanism. The repointer shall be constructed so that it can be mounted on a desk or table top by means of a clamp with a minimum opening of 3 inches designed in accordance with commercial practice for adjustments to various widths.

3.3 Construction. The construction shall be sufficiently substantial to assure serviceability and durability. The central spindle and the top shall have their surfaces constructed in such a manner as to permit smooth and easy operation. The interchangeable bushings shall fit smoothly and snug into the aperture and be removable without tools.

3.4 Performance. The repointer shall be capable of pointing lead to hairline points and with short to long tapers without breaking the leads. The resulting leads shall be cone-shaped without flat surfaces. (See 4.4.2.)

3.5 Finish. All exterior surfaces shall be finished in accordance with the best commercial practice. Exposed metal surfaces, other than corrosion resistant metal surfaces and cutting blades or drums, shall have a corrosion inhibiting coating. Plastic surfaces shall be smooth and without imperfections which would impair serviceability or appearance.

3.6 Repointing elements. The repointing elements shall be available for procurement as replacements.

3.7 Product marking. Each lead repointer shall be plainly and permanently marked with the manufacturer's name or readily determined trade mark.

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3.8 Workmanship. The repointers and parts thereof shall be neatly made, smoothly finished, and all moving parts shall function satisfactorily. Repointers shall be completely without fault which might constitute a personal hazard to the user.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For the purpose of sampling a lot shall consist of all lead repointers of one type offered for delivery at one time as defined in MIL-STD-105.

4.2.2 Sampling for visual examination. A random sample of lead repointers shall be taken from each lot in accordance with MIL-STD-105, inspection level S-5. The acceptable quality level (AQL) shall be 2.5 percent defective. The sample unit shall be one lead repointer.

4.2.3 Sampling for test. A random sample of lead repointers shall be taken from each lot in accordance with MIL-STD-105, inspection level S-4. The AQL shall be 2.5 percent defective. The sample unit shall be one lead repointer.

4.2.4 Sampling for preparation for delivery. Sampling shall be in accordance with MIL-STD-105. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. The inspection level shall be level II and the Acceptable Quality Level (AQL) shall be 6.5 defects per hundred units.

4.3 Visual examination. Samples taken in accordance with 4.2.2 shall be examined for defects listed in Table I, and other defects of design, construction or appearance. Any lead repointers in the sample containing one or more defects shall be rejected. If the number of defective lead repointers exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE I. - Visual examination defects

Material	Not of good grade. Not appropriate to the component and its functions. Not such as to assure good appearance. Not such as to assure good serviceability.
Design	Does not permit one-handed use. Does not operate by insertion of lead thru aperture in top of repointer and revolving of top. Not specified type. Abrasive medium not readily replaceable. Top not readily removable. Pointing medium not replaceable without special tools.
Construction	Not durable. Not sufficiently substantial to assure serviceability. Bottom does not have pad that is adherent.
Finish	Coating not of good quality or not of good appearance. Exposed metal surfaces do not have corrosion-inhibiting coating, when required. Plastic finishes not smooth.
Workmanship	Not devoid of burrs, slivers, sharp edges or other faults which might be a personal hazard. Cracks, chips or faults which might impair serviceability. Faults adversely affecting appearance.
Product Marking	Not plainly marked with manufacturer's name or trademark.

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4.4 Test. Samples taken in accordance with 4.2.3 shall be tested as described in 4.4.1 and 4.4.2. Any lead repointers in the sample containing one or more defects shall be rejected. If the number of defective lead repointers in the sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.1 Hardness test. The drum or cutter shall be tested in accordance with ASTM E18.

4.4.2 Performance test. The lead repointer shall be placed or clamped on a horizontal surface comparable to a draftsman's drawing board and a graphite lead of 2H hardness of plastic lead normally used for this purpose shall be placed in a mechanical drawing pencil. The lead shall be pointed three times, with the point broken off between pointing action and observed for compliance with the requirements of 3.4. The repointing operation shall be performed using only one hand and in a normal manner. The pointing process shall be observed for compliance with 3.4 and to assure that the top revolves smoothly and easily as required by 3.3. Repointer shall also be capable of pointing woodcased drawing pencils after the wood casing has been removed an appropriate distance to permit pointing the lead.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with Table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

4.6 The sum of defectives in 4.3 and 4.4 in the aggregate shall not exceed AQL 2.5 at level S4.

TABLE II. - Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.
Contents	Net weight exceeds the requirements of the applicable box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Each lead repointer shall be individually packaged in a close-fitting box, conforming to PPP-B-566 or PPP-B-676, style and class at the option of the contractor. Each box shall be waterproofed in accordance with MIL-P-116, method IC-2.

5.1.2 Level B. Each repointer shall be packaged as specified in 5.1 except that waterproofing shall not be required.

5.1.3 Level C. Repointers shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified.

5.2.1 Level A. Unless otherwise specified, twelve (12) lead repointers of one type, packaged as specified in 5.1.1, shall be packed in a close-fitting box conforming to PPP-B-636, V3c or V3s, style RSC. The container shall be closed and sealed in accordance with the appendix to the box specification.

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5.2.2 Level B. Twelve (12) lead repointers of one type, packaged as specified in 5.1.2, shall be packed in a close-fitting box conforming to PPP-B-636, style RSC, minimum bursting strength of 200 p.s.i. The boxes shall be closed and sealed in accordance with method II of the appendix to the box specification.

5.2.3 Level C. The repointers shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the National Motor Freight Classification Rules and the Uniform Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lead repointers covered by this specification are intended for use by draftsmen and artists. They are intended for one-handed use in pointing leads held in mechanical lead holders and leads which are wood encased (after the wood casing has been cut back an appropriate distance to permit pointing the lead).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type (see 1.2.1).
(c) Inspection responsibility, if other than as specified herein (see 4.1).
(d) Levels of packaging and packing and any special marking required (5.1., 5.2., and 5.3).
(e) Whether fine or coarse grit is desired for type I (see 3.2.1).

User Activity:

Army - GL

Civil Agency Coordinating Activity:

GSA-FSS

Preparing Activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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